

GINGIN, Australia

WMO: 956120

Lat: 31.463S

Lon: 115.864E

Elev: 244

StdP: 14.57

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.8	37.2	26.3	20.5	67.6	29.8	24.1	62.5	23.1	59.9	21.0	59.2	2.7	130	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.4	99.7	66.9	96.1	66.3	92.8	66.0	72.1	86.3	70.5	85.0	69.1	83.7	13.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	105.7	75.5	66.6	98.6	74.2	64.8	92.8	73.2	36.0	86.3	34.5	84.9	33.4	84.0	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	20.7	18.9	DB	30.0	106.7	2.1	3.0	28.5	108.8	27.2	110.6	26.1	112.2	24.5	114.4	(4)
(5)			WB	29.6	75.0	2.6	1.8	27.7	76.3	26.2	77.3	24.8	78.3	22.9	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	75.8	75.9	72.9	66.2	60.0	55.6	53.4	54.7	56.4	61.2	67.6	71.2		
	DBStd	9.88	6.48	6.45	6.79	6.07	5.18	4.34	4.37	4.13	4.41	5.72	6.72	7.12		
	HDD50	33	0	0	0	0	1	6	18	6	2	0	0	0		
	HDD65	1677	2	3	11	56	169	284	359	319	261	150	46	17		
	CDD50	5211	799	726	709	487	312	173	124	153	195	347	529	657		
	CDD65	1378	336	309	255	93	15	1	0	3	32	125	209			
	CDH74	15199	3742	3380	2628	866	173	5	0	12	77	427	1461	2428		
	CDH80	7183	1961	1705	1210	279	28	0	0	1	13	130	646	1210		
(14) Wind	WSAvg	9.3	11.5	11.2	10.8	8.4	7.4	7.5	7.5	8.0	8.5	9.4	10.4	11.1		
(15) Precipitation	PrecAvg	26.3	0.6	0.6	0.8	1.3	3.5	4.8	5.2	4.3	2.7	1.4	0.8	0.4		
	PrecMax	34.2	4.0	3.1	3.4	2.9	6.6	9.2	8.1	6.5	5.4	3.5	2.5	1.8		
	PrecMin	17.4	0.0	0.0	0.0	0.2	1.7	1.9	2.2	0.9	0.6	0.5	0.1	0.0		
	PrecStd	3.6	1.1	0.7	0.8	0.7	1.4	1.8	1.5	1.3	1.1	0.7	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.5	102.6	101.1	92.1	83.3	74.3	70.9	75.3	81.2	90.1	98.1	102.8		
		MCWB	66.6	68.5	67.2	63.5	59.5	56.0	56.5	57.4	59.4	62.8	63.3	65.4		
	2%	DB	98.8	97.8	95.5	86.8	78.7	71.0	67.5	70.9	75.4	83.2	92.5	97.3		
		MCWB	66.8	67.4	66.9	63.6	59.2	57.0	56.1	57.5	58.5	61.4	63.1	65.3		
	5%	DB	94.8	94.2	90.9	82.9	74.8	68.0	64.9	67.2	70.9	78.3	87.4	92.0		
		MCWB	66.9	67.0	66.1	63.0	58.4	56.2	55.9	56.4	57.5	60.6	62.7	64.9		
	10%	DB	90.7	90.3	86.8	78.9	71.4	65.5	62.7	64.5	67.3	74.0	82.5	86.7		
		MCWB	66.4	66.9	65.4	62.4	58.4	55.8	55.1	55.9	56.7	59.6	62.0	63.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.8	74.6	73.8	69.5	66.8	63.7	62.4	63.3	63.4	66.4	68.8	70.5		
		MCDB	87.5	89.7	86.6	79.0	72.7	67.6	64.8	67.4	72.0	78.7	84.1	91.4		
	2%	WB	71.8	72.3	71.0	67.7	64.3	61.4	59.5	60.8	61.3	64.1	66.9	68.6		
		MCDB	85.9	86.8	84.0	77.4	70.0	65.7	63.4	65.2	69.5	77.5	82.2	87.4		
	5%	WB	70.2	70.2	69.3	66.2	62.4	59.5	57.7	58.9	59.6	62.5	65.4	67.1		
		MCDB	84.6	85.9	81.7	75.8	69.3	64.0	61.8	64.3	67.2	73.6	79.6	83.9		
	10%	WB	68.6	68.7	67.6	64.6	60.6	57.6	56.2	57.1	58.1	61.0	63.9	65.7		
		MCDB	83.5	84.1	80.3	74.9	67.8	62.6	60.6	62.5	65.2	71.5	77.0	80.7		
(35) Mean Daily Temperature Range	MDBR		26.1	25.4	24.6	24.0	23.0	20.6	19.8	20.3	22.0	24.4	25.5	25.7		
	5% DB	MCDBR	32.3	29.9	29.6	28.7	28.4	25.1	22.3	26.1	29.8	30.7	32.9	33.9		
		MCWBR	9.9	9.3	10.1	11.5	13.6	14.3	13.6	14.7	15.4	13.2	11.3	10.6		
	5% WB	MCDBR	25.3	25.5	22.6	22.3	20.1	18.2	16.8	18.7	23.6	25.8	27.6	28.0		
		MCWBR	9.2	9.6	9.6	11.0	12.5	12.8	11.8	12.2	13.7	12.4	11.2	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.341	0.325	0.316	0.306	0.296	0.297	0.303	0.324	0.325	0.326	0.328		
	taud		2.563	2.573	2.609	2.620	2.617	2.631	2.620	2.601	2.517	2.533	2.555	2.568		
	Ebn at Noon		317	311	305	292	279	275	280	292	300	312	319	322		
	Edh at Noon		34	33	30	26	23	22	23	26	32	34	34	34		
(44) All-Sky Solar Radiation	RadAvg		2612	2359	1959	1449	1091	899	949	1224	1627	2103	2489	2678		
	RadStd		117	76	96	81	72	63	57	56	89	89	94	108		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page